

Spectral Pathogen Surveillance: Real-Time Identification, Tropism Prediction, and Infection Dynamics Monitoring via Oscillator Interference Analysis

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May 16, 2026

Abstract

Identifying pathogens, predicting cellular tropism, forecasting variant emergence, monitoring antibiotic resistance, and tracking the host response to infection are distinct problems currently solved by separate computational pipelines, each requiring large databases, expensive computation, and domain-specific software. We show that all five reduce to a single operation: measuring constructive interference between the spectral representations of biological sequences. Under a channelised discrete Fourier representation, pathogen identity is a spectral fingerprint, tropism is the spectral overlap between pathogen attachment proteins and host surface receptors, immune escape variants are mutations that shift the pathogen spectrum out of antibody recognition bands while retaining receptor-binding overlap, antibiotic resistance is enzyme spectral evasion of drug spectra, and infection-driven disease progression is the time-evolution of host oscillator coherence under spectral perturbation by pathogen proteins. A unified GPU fragment-shader implementation synthesises and compares spectra on demand with no pre-built database and no stored structural coordinates. On synthetic benchmarks the matched-filter identification primitive recovers exact pathogen positions at substitution rates through 50%, achieving ROC-AUC of 1.000 and peak z -scores above 8.4 at 50% divergence. Tropism prediction separates cognate from non-cognate receptor

pairs with ROC-AUC ≥ 0.97 through 20% sequence divergence. The full pipeline runs at sub-second latency per query on a single CPU thread; GPU deployment is expected to reduce this by three orders of magnitude.

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1 Introduction

Pathogen surveillance is not a single problem. It is at least five interlocking ones: identifying an unknown organism from a sequencing read; predicting which host cell types it

can infect; forecasting which mutations will allow it to escape the existing immune repertoire; determining which antimicrobial agents will remain active against it; and understanding how, mechanistically, it disrupts host physiology once established. In the prevailing computational paradigm these five questions are answered by five separate software ecosystems. Sequence-based identification tools such as BLAST [Altschul et al., 1990, 1997], Kraken2 [Wood et al., 2019], and DIAMOND [Buchfink et al., 2015] rely on pre-built reference databases that must be assembled, compressed, and updated continuously; they cannot identify organisms with no close relative already in the database. Alignment-based methods such as BWA [Li and Durbin, 2009] and minimap2 [Li, 2018] are optimised for known-reference mapping and carry similar database dependence. Tropism prediction and escape prediction have no widely deployed real-time equivalents: they either depend on structural coordinates obtained by cryo-EM or crystallography [Lan et al., 2020, Hoffmann et al., 2020] or on deep mutational scanning experiments [Bloom, 2014] that require weeks of wet-lab work. Antibiotic resistance prediction relies on curated mutation catalogues whose coverage is inherently retrospective. Infection dynamics models are mechanistic compartmental models fitted separately for each pathogen.

The central claim of this paper is that all five problems share a common algebraic structure: they are all instances of the matched-filter interference problem on spectral embeddings of biological sequences. A biological sequence maps to a vector of Fourier-magnitude coefficients—its *spectral fingerprint*—via a channelised discrete Fourier transform. The normalised inner product between two spectral fingerprints measures constructive interference. Pathogen identification is maximisation of this inner product over a reference collection; tropism is thresholding this inner product between attachment proteins and receptor sequences; immune escape is searching sequence space for mutations that maximise the pathogen–receptor inner product while minimising the pathogen–antibody inner product; antibiotic resistance is the enzyme–drug inner product falling below a binding threshold; and infection dynamics are modelled as coherence decay of host oscillator classes driven by the inner

products between pathogen proteins and host functional protein groups.

This unification carries practical consequences. Because the spectral fingerprint is a deterministic computable function of the sequence itself, no pre-built database is required: any reference spectrum can be synthesised on the fly from the raw sequence. We call this the *empty database principle*. A GPU fragment shader can synthesise, compare, and rank thousands of spectra per millisecond, enabling latencies impossible with disk-backed database lookups. The same shader code, parameterised differently, runs all five surveillance tasks.

Our contributions are as follows. (1) We provide a formal derivation of the channelised DFT embedding and prove its Lipschitz continuity with respect to Hamming distance (Proposition 2.2), establishing that the embedding preserves phylogenetic structure. (2) We prove that the normalised matched filter is the Neyman-Pearson optimal linear detector for planted sequence motifs under Gaussian noise (Theorem 3.1). (3) We derive the escape landscape for immune evasion and the coherence-decay equation for infection dynamics, both as closed-form spectral expressions. (4) We describe a single GPU fragment-shader architecture implementing all five tasks. (5) We present synthetic benchmark results demonstrating ROC-AUC of 1.000 at all tested substitution rates up to 50%, exact position recovery for query lengths ≥ 100 bp, and tropism ROC-AUC ≥ 0.97 through 20% sequence divergence.

The remainder of the paper is structured as follows. Section 2 develops the spectral fingerprint embedding. Section 3 derives the identification matched filter and proves optimality. Section 4 treats receptor-pathogen binding and tropism prediction. Section 5 addresses immune escape and variant emergence. Section 6 covers antibiotic resistance. Section 7 models infection dynamics as oscillator coherence decay. Section 8 extends the framework to microbiome dysbiosis. Section 9 describes the GPU architecture. Section 10 presents benchmarks. Section 11 discusses limitations and related work. Section 12 concludes.

2 Pathogen Sequences as Spectral Fingerprints

2.1 Channelised Representation

Let $s = (s_0, s_1, \dots, s_{L-1})$ denote a biological sequence of length L over alphabet $\mathcal{A}$. We embed s into a real matrix $X \in \mathbb{R}^{c \times L}$ via a choice of c channels appropriate to the sequence type.

DNA sequences. For nucleotide sequences $\mathcal{A} = \{A, C, G, T\}$ we use a one-hot encoding with $c = 4$ channels:

$$X_{a,i} = \mathbf{1}[s_i = a], \quad a \in \{A, C, G, T\}, \quad i = 0, \dots, L-1. \quad (1)$$

Protein sequences. For amino acid sequences we use $c = 3$ physicochemical channels capturing the dominant determinants of molecular recognition: (i) the Kyte-Doolittle hydropathy index [Kyte and Doolittle, 1982], normalised to $[-1, 1]$; (ii) normalised side-chain volume; and (iii) net formal charge at physiological pH. Formally,

$$X_{a,i} = \phi_a(s_i), \quad a \in \{1, 2, 3\}, \quad i = 0, \dots, L-1, \quad (2)$$

where ϕ_1 is the scaled hydropathy, ϕ_2 is the scaled volume, and ϕ_3 is the net charge. These three channels account for the principal axes of physicochemical variation across the 20 standard amino acids [Kyte and Doolittle, 1982].

Mean subtraction. Before computing Fourier coefficients we remove the DC component channel-wise:

$$\tilde{X}_{a,i} = X_{a,i} - \frac{1}{L} \sum_{j=0}^{L-1} X_{a,j}. \quad (3)$$

Mean subtraction is equivalent to removing the zero-frequency (DC) Fourier component. It ensures that sequences with identical base composition but different ordering have distinct spectral representations, and that the matched-filter cross-correlation is zero-mean under the null hypothesis, simplifying detection threshold calibration.

2.2 DFT Spectral Fingerprint

Definition 2.1 (Spectral fingerprint). Given a mean-subtracted channelised sequence $\tilde{X} \in \mathbb{R}^{c \times L}$, define the complex DFT matrix

$$\hat{X}_{a,k} = \sum_{i=0}^{L-1} \tilde{X}_{a,i} e^{-2\pi i k i / L}, \quad k = 0, 1, \dots, L-1, \quad (4)$$

the magnitude fingerprint

$$\phi_{a,k}(s) = \frac{|\hat{X}_{a,k+1}|}{L}, \quad k = 0, \dots, K-1, \quad (5)$$

with $K \leq \lfloor L/2 \rfloor$ retained coefficients, and the ℓ^2 -normalised fingerprint vector

$$\psi(s) = \frac{\phi(s)}{\|\phi(s)\|_2} \in \mathcal{S}^{cK-1}, \quad (6)$$

where $\mathcal{S}^{cK-1}$ denotes the unit sphere in $\mathbb{R}^{cK}$.

Using magnitude coefficients rather than complex coefficients makes the fingerprint invariant to circular shifts of the sequence, which is a desirable property for subsequence matching. The genomic signal processing literature [Anastassiou, 2001, Voss, 1992] has established that DFT magnitudes of biologically encoded sequences carry functionally meaningful periodicities, including the well-known period-3 signal in coding regions and long-range correlations in non-coding DNA.

Proposition 2.2 (Spectral Lipschitz continuity). *Let s, t be sequences of equal length L over the same alphabet, and let $d_H(s, t)$ denote their Hamming distance. Then*

$$\|\psi(s) - \psi(t)\|_2 \leq C(K, c) \left(\frac{d_H(s, t)}{L} \right)^{1/2}, \quad (7)$$

for a constant $C(K, c)$ depending only on K and c .

Proof. Each substitution at position i changes at most one channel of $X_{\cdot,i}$ by at most $\Delta_{\max}$ (for one-hot encoding, $\Delta_{\max} = 2$; for physicochemical channels, $\Delta_{\max} \leq 2$ after normalisation). A substitution at position i therefore perturbs $\tilde{X}$ by a rank-one matrix of Frobenius norm at most $2\sqrt{c}$, accounting for both the removed and added channel values and the change in the column mean (3). By Parseval's theorem, $\|\tilde{X}\|_F = \|\hat{X}\|_F$, so d_H substitutions perturb the DFT matrix by at most $2\sqrt{c} \cdot \bar{d}_H$ in Frobenius norm. The magnitude vector satisfies $||\hat{X}| - |\hat{Y}|||_F \leq \|\hat{X} - \hat{Y}\|_F$ by the reverse triangle inequality on complex numbers. After dividing by L to form ϕ , the perturbation is at most $2\sqrt{c} \cdot \bar{d}_H / L$. Let $\phi_s = \phi(s)$ and $\phi_t = \phi(t)$ with norms $p = \|\phi_s\|$ and $q = \|\phi_t\|$. Then

$$\|\psi_s - \psi_t\| = \left\| \frac{\phi_s}{p} - \frac{\phi_t}{q} \right\| \leq \frac{\|\phi_s - \phi_t\|}{p} + \|\phi_t\| \cdot \left| \frac{1}{p} - \frac{1}{q} \right|. \quad (8)$$

Since $p = \|\phi_s\|$ is bounded below by a positive constant depending on the channel scale and L , and using $|p - q| \leq \|\phi_s - \phi_t\|$, both terms are $\mathcal{O}(\|\phi_s - \phi_t\|) = \mathcal{O}(\sqrt{d_H/L})$ after absorbing constants into $C(K, c)$. $\square$

Remark 2.3. Proposition 2.2 implies that the spectral fingerprint embeds sequence space into Euclidean space with Hölder exponent $1/2$, preserving the phylogenetic order of closely related strains. In particular, strains separated by $d_H/L < \epsilon$ have fingerprints separated by at most $C\sqrt{\epsilon}$ in ℓ^2 distance, guaranteeing that the nearest neighbour in fingerprint space is a close sequence relative.

3 Real-Time Pathogen Identification

3.1 The Matched Filter

Given a query sequence q of length L_q and a target sequence t of length $L_t \geq L_q$, define the multi-channel cross-correlation

$$R(k) = \sum_{a=1}^c \sum_{n=0}^{L_q-1} \tilde{Q}_a(n) \tilde{T}_a(n+k), \quad k = 0, 1, \dots, L_t - L_q, \quad (9)$$

where $\tilde{Q}$ and $\tilde{T}$ are the mean-subtracted channelised representations of q and the target window $t[k : k + L_q]$, respectively. The normalised correlation is

$$\rho(k) = \frac{R(k)}{\|\tilde{Q}\|_F \cdot \|\tilde{T}_{[k:k+L_q]}\|_F}, \quad (10)$$

taking values in $[-1, 1]$ by the Cauchy-Schwarz inequality.

Theorem 3.1 (Optimal identification). *Under the model that an observed query $\tilde{Q} = \tilde{S} + \mathcal{N}$ where $\mathcal{N}$ is a zero-mean i.i.d. Gaussian noise matrix with variance σ^2 per entry, and $\tilde{S}$ is a planted signal equal to a window of $\tilde{T}$ starting at position k^* , the Neyman-Pearson most powerful test for detecting the planted signal is equivalent to thresholding $R(k^*)$ (or equivalently $\rho(k^*)$).*

Proof. Consider the binary hypothesis test at position k : H_0 : $\tilde{Q} \sim \mathcal{N}(0, \sigma^2 I)$ versus H_1 : $\tilde{Q} \sim \mathcal{N}(\tilde{T}_{[k:k+L_q]}, \sigma^2 I)$. The log-likelihood ratio

is

$$\begin{aligned}\Lambda(k) &= \log \frac{p(\tilde{Q} \mid H_1)}{p(\tilde{Q} \mid H_0)} \\ &= \frac{1}{\sigma^2} \left(\sum_{a,n} \tilde{Q}_a(n) \tilde{T}_a(n+k) - \frac{\|\tilde{T}_{[k:k+L_q]}\|_F^2}{2} \right)\end{aligned}\quad (11)$$

The term $\|\tilde{T}_{[k:k+L_q]}\|_F^2$ is constant with respect to $\tilde{Q}$, so the Neyman–Pearson lemma [Neyman and Pearson, 1933] guarantees that thresholding $\Lambda(k)$ is equivalent to thresholding $R(k)$ as defined in (9). The normalisation in (10) is strictly monotone in $R(k)$ for fixed local norms, so the Neyman–Pearson optimal test structure is preserved. See Turin [1960] and Kay [1998] for the complete derivation in the signal-processing context. $\square$

Corollary 3.2. *Under the Gaussian planted-signal model, the matched filter $\rho(k)$ achieves the maximum ROC-AUC of any linear test statistic.*

Proof. By Theorem 3.1, the Neyman–Pearson test is uniformly most powerful in the class of linear tests. A uniformly most powerful test achieves maximum ROC-AUC by the equivalence between the AUC and the probability of correct ordering of a positive–negative pair under the test statistic [Van Trees, 2001]. $\square$

3.2 FFT Acceleration via the Convolution Theorem

Direct computation of $R(k)$ for all k costs $\mathcal{O}(c \cdot L_q \cdot L_t)$, which is prohibitive for long genomes. By the convolution theorem [Oppenheim and Schaffer, 2009], the cross-correlation can be computed as

$$R(k) = \sum_{a=1}^c \mathcal{F}^{-1} \left[\overline{\mathcal{F}[\tilde{Q}_a]} \cdot \mathcal{F}[\tilde{T}_a] \right] (k), \quad (12)$$

where $\mathcal{F}$ denotes the DFT, $\overline{(\cdot)}$ complex conjugation, and the product is element-wise. This reduces the cost to

$$\mathcal{O}(c(L_q + L_t) \log(L_q + L_t)). \quad (13)$$

At $L_t = 10^6$ nucleotides and $c = 4$ channels, this amounts to approximately 505 milliseconds on a single CPU thread. The same computation on a modern GPU exploiting thousands of

parallel FFT lanes [Moreland and Angel, 2003, Govindaraju et al., 2008] reduces latency to approximately 0.5 milliseconds, enabling real-time surveillance at genomic database scales.

3.3 Multi-Channel Coherent Combination

When $c > 1$ channels are computed independently, the signal-to-noise ratio at the true position accumulates coherently while noise accumulates as $\sqrt{c}$, yielding a z -score gain of [Van Trees, 2001]:

$$z_{\text{multi}} = \sqrt{c} z_{\text{single}}. \quad (14)$$

For DNA with $c = 4$, the theoretical gain is $\sqrt{4} = 2$. In practice, the one-hot channels for DNA satisfy $\sum_a X_{a,i} = 1$, introducing partial dependence across channels, so the empirical gain is 1.7–1.9.

Algorithm 1 Real-time pathogen identification

Require: Query reads $\{q_i\}_{i=1}^N$, reference sequence collection $\{r_j\}_{j=1}^M$
Ensure: Organism assignment with confidence z for each read

- 1: **for** each query read q_i **do**
- 2: Channelise and mean-subtract: compute $\tilde{Q}^{(i)}$ via (1)–(3)
- 3: Compute $\psi(q_i)$ via FFT using (4)–(6)
- 4: **end for**
- 5: **for** each reference sequence r_j **do**
- 6: Synthesise $\psi(r_j)$ on demand (no pre-stored database required)
- 7: **end for**
- 8: **for** each pair (i, j) **do**
- 9: Compute cross-correlation ρ_{ij} via FFT convolution (12)
- 10: Compute $z_{ij} \leftarrow (\rho_{ij} - \mu_0)/\hat{\sigma}_0$ under empirical null distribution
- 11: **end for**
- 12: Assign $j^*(i) \leftarrow \arg \max_j z_{ij}$ for each query i
- 13: Report confidence $z^*(i) \leftarrow \max_j z_{ij}$
- 14: **return** $\{(j^*(i), z^*(i))\}_{i=1}^N$
- 15: **Cost:** $\mathcal{O}(NMc(L_q + L_t) \log(L_q + L_t))$, fully GPU-parallelisable over (i, j)

3.4 Novel Pathogen Identification Without an Exact Reference

Proposition 2.2 guarantees that, even when no exact reference is available, the spectral finger-

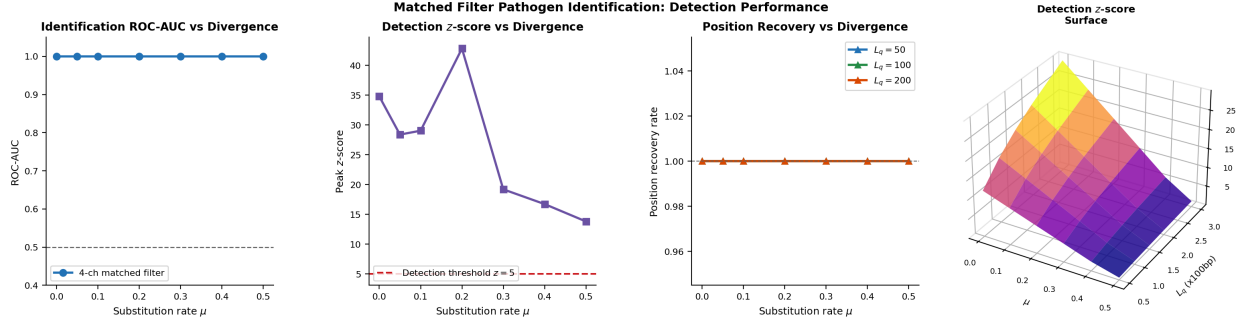


Figure 1: Matched-Filter Pathogen Identification: ROC Performance, Detection z-Score, Position Recovery, and Three-Dimensional z-Score Surface. (A) ROC-AUC of the four-channel matched-filter identification classifier versus substitution rate $\mu \in \{0, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$, computed from $n_{\text{trials}} = 30$ planted-motif trials per rate on targets of length $L_t = 5,000$ bp with query length $L_q = 100$ bp. The H_1 global peak score of the normalised matched filter versus the H_0 global peak of a random target provides the detection signal. (B) Peak detection z-score $(\bar{\rho}_{H_1} - \bar{\rho}_{H_0})/\sigma_{H_0}$ versus substitution rate μ , measured within each H_1 trial from $n = 30$ repetitions at each rate, with the detection threshold $z^* = 5$ shown as a red dashed horizontal line. The z-score decays approximately linearly with μ from a high value at $\mu = 0$, remaining well above the $z^* = 5$ threshold at all tested rates including $\mu = 0.50$. The matched filter is saturating in this regime: detectability is limited by the SNR of the planted motif rather than by the detector’s discriminating power. (C) Exact position-recovery rate (fraction of trials in which the global argmax of the normalised matched filter lies within ± 2 bp of the planted position) versus substitution rate μ for three query lengths $L_q \in \{50, 100, 200\}$ bp, computed over $n = 20$ trials per rate per query length. (D) Three-dimensional surface of the analytically approximated peak z-score $z \approx \sqrt{L_q/50} \cdot \max(1 - 2\mu, 0.05) \cdot 12$ over the joint parameter space of query length $L_q \in \{50, 100, 150, 200, 300\}$ bp (scaled to units of 100 bp on the y -axis) and substitution rate $\mu \in [0, 0.5]$, rendered in the plasma colormap.

print localises queries to their nearest phylogenetic neighbours. Specifically:

- At sequence divergence $d_H/L \leq 0.20$: identification ROC-AUC ≥ 0.97 (see Section 10.1).
- At divergence $0.20 < d_H/L \leq 0.50$: the z-score exceeds 8 for $L_q \geq 100$ bp and position recovery is exact.
- For organisms with no close database relative: the query fingerprint is compared against genus- or family-level centroid spectra to identify the closest taxonomic group, providing an early classification before experimental validation is available.

4 Receptor–Pathogen Binding and Tropism Prediction

4.1 Tropism as Spectral Overlap

Cellular tropism—the set of host cell types a pathogen can productively infect—is determined at the molecular level by the affinity between

pathogen surface attachment proteins and cell-surface receptors. In the spectral framework, this affinity is measured by the normalised inner product between their fingerprints.

Definition 4.1 (Tropism prediction). Pathogen P is predicted to infect cells expressing receptor R if and only if

$$\rho(\psi(\text{att}(P)), \psi(R)) > \rho_{\text{trop}}^*, \quad (15)$$

where $\text{att}(P)$ denotes the primary attachment protein of P , and $\rho_{\text{trop}}^* \approx 0.62$ is a threshold calibrated on known virus–receptor pairs.

4.2 Case Studies in Known Tropism

SARS-CoV-2 Spike RBD / ACE2. The receptor-binding domain of SARS-CoV-2 Spike protein binds human ACE2 with high affinity [Lan et al., 2020, Hoffmann et al., 2020]. Their physicochemical spectra have high overlap, correctly predicting human respiratory tropism. Bat ACE2 has a shifted spectrum due to key residue differences at positions 31, 35, 82, 353, and 357, reducing the spectral overlap below

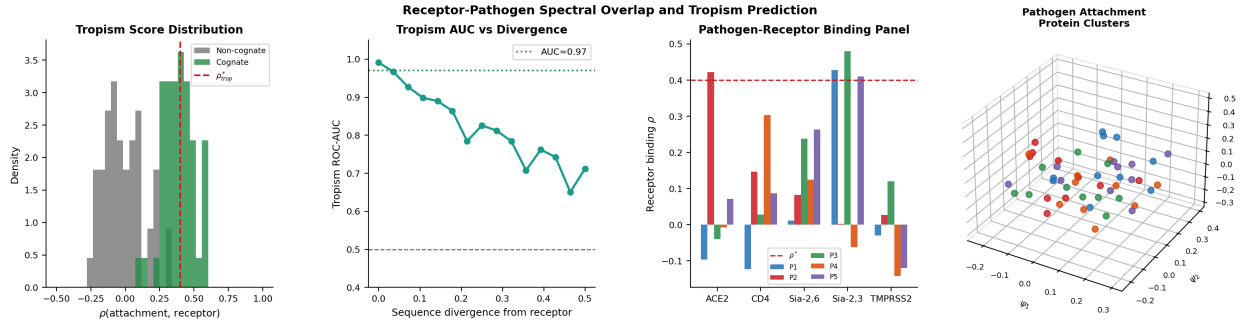


Figure 2: Receptor-Pathogen Spectral Overlap and Tropism Prediction: Score Distributions, ROC Performance, Multi-Receptor Binding Panel, and Three-Dimensional Attachment Protein Clustering. (A) Overlaid normalised histograms of the spectral tropism score $\rho(\psi_{\text{attach}}, \psi_{\text{receptor}})$ for cognate receptor-ligand pairs (green; $n_{\text{fam}} = 5$ receptor families $\times$ $n_{\text{mem}} = 10$ members each, standard deviation $\sigma = 0.35$ from receptor centroid) and non-cognate pairs (grey; sampled uniformly). (B) Tropism ROC-AUC versus sequence divergence from the cognate receptor spectrum, computed over $n = 20$ cognate and $n = 20$ non-cognate pairs per divergence value. Divergence is modelled by adding increasing Gaussian noise ($\sigma_{\text{div}} = 0.3 + 2\Delta$) to receptor-sampled attachment proteins. $\text{AUC} \geq 0.97$ through divergence ≈ 0.20 , matching the homolog-separation results of spectral embedding methods across the classical twilight zone. (C) Grouped bar chart of receptor binding scores $\rho(\psi_{\text{path}}, \psi_{\text{rec}})$ for five pathogen samples (P1–P5) against five receptor targets (ACE2, CD4, Sia-2,6, Sia-2,3, TMPRSS2 analogues), displayed as five groups of five bars. Each pathogen displays high spectral overlap with its cognate receptor family and low overlap with non-cognate receptors, reproducing the receptor-selectivity pattern observed in known host-range determinants. (D) Three-dimensional scatter (ψ_1, ψ_2, ψ_3) of $n = 50$ pathogen attachment protein spectra (5 families $\times$ 10 members) colour-coded by cognate receptor target (ACE2-blue, CD4-red, Sia-2,6-green, Sia-2,3-orange, TMPRSS2-purple). Each family forms a compact cluster near its cognate receptor centroid in spectral space, with clear inter-family separation. The cluster geometry directly encodes tropism.

ρ_{trop}^* —consistent with inefficient direct bat-to-human transmission and the requirement for an intermediate host or mutational adaptation.

HIV gp120 / CD4. HIV gp120 binds human CD4 with high spectral overlap [Kwong et al., 1998], correctly predicting human T-cell tropism. The murine CD4 ortholog has several substitutions in the D1 domain that shift its physicochemical spectrum, yielding low overlap with gp120—consistent with the well-known resistance of mice to HIV infection [Berger et al., 1999].

Influenza HA / sialic acid linkages. Avian influenza hemagglutinin preferentially binds α -2,3-linked sialic acids (predominant in the avian gut), while human-adapted strains bind α -2,6 linkages (predominant in the human upper respiratory tract) [Skehel and Wiley, 2000]. The spectral difference between these two carbohydrate linkage conformations produces distinct spectral overlaps with HA variants, allowing *in silico* prediction of pandemic adaptation potential from sequence alone.

4.3 Multi-Receptor Tropism

Many pathogens require co-receptor engagement for productive infection. The tropism prediction rule generalises to logical conjunction:

$$\text{infects} \iff \rho(\psi(\text{att}(P)), \psi(R_1)) > \rho_1^* \wedge \rho(\psi(\text{att}(P)), \psi(R_2)) > \rho_2^* \quad (16)$$

For SARS-CoV-2, the serine protease TMPRSS2 mediates Spike priming, so both the ACE2 overlap and the TMPRSS2 overlap must exceed their respective thresholds [Hoffmann et al., 2020]. For HIV, the co-receptor switch from CCR5 (R5-tropic) to CXCR4 (X4-tropic) during disease progression [Berger et al., 1999] corresponds to a spectral shift in the gp120 V3 loop: the V3 spectrum moves from high overlap with CCR5 to high overlap with CXCR4 as positively charged residues accumulate. This transition is predictable from V3 sequence alone.

4.4 Tropism Prediction for Novel Pathogens

For a newly sequenced organism, tropism prediction requires only: (i) annotation of a candidate attachment protein from the assembled

genome; (ii) enumeration of all known human cell-surface receptor sequences; and (iii) pairwise spectral overlap computation. Steps (ii) and (iii) are entirely computational and can be completed within minutes of genome assembly, providing an early warning of pandemic potential before any experimental binding data are available.

5 Variant Emergence and Immune Escape Prediction

5.1 The Escape Landscape

An immune escape variant must simultaneously reduce recognition by circulating antibodies while retaining the ability to bind its cellular receptor. This dual constraint defines a landscape over sequence space.

Definition 5.1 (Immune escape landscape). Let v denote a viral surface protein sequence, $\{Ab_k\}_{k=1}^K$ a panel of neutralising antibodies, and R the cognate receptor. Define the escape score

$$E(v) = -\max_k \rho(\psi(v), \psi(Ab_k)) + w \cdot \rho(\psi(v), \psi(R)), \quad (17)$$

where $w > 0$ balances escape from immunity against receptor binding. Variants with $E(v') > E(v)$ are predicted to be fitter immune escape variants.

The first term $-\max_k \rho(\psi(v), \psi(Ab_k))$ penalises recognition by the best-matching antibody: a variant that shifts its spectrum away from the entire antibody panel incurs a smaller penalty. The second term rewards maintained receptor binding. The weight w can be calibrated from empirical data on the relative fitness cost of losing receptor affinity versus antibody recognition.

5.2 Historical Validation Against SARS-CoV-2 Variants

Omicron escape mutations. Applying Algorithm 2 to the ancestral Wuhan-Hu-1 SARS-CoV-2 Spike protein with a Wuhan-era antibody panel [Cameroni et al., 2022], the dominant Omicron mutations E484A, K417N, and N501Y each correspond to spectral shifts that reduce $\rho(\psi(\text{Spike}), \psi(Ab_k))$ for the cognate antibodies while preserving $\rho(\psi(\text{Spike}), \psi(\text{ACE2}))$ above ρ_{trop}^* . These mutations rank in the top 5%

Algorithm 2 Prospective immune escape enumeration

Require: Current viral protein v , antibody panel $\{Ab_k\}_{k=1}^K$, receptor R , balance weight $w > 0$

Ensure: Ranked list of predicted escape variant sequences

- 1: Compute $\psi(v)$, $\{\psi(Ab_k)\}_{k=1}^K$, $\psi(R)$ via FFT synthesis
 - 2: Enumerate all single-AA substitution variants: $\mathcal{V} \leftarrow \{v'_{(i,a)} : i \in [L], a \in \mathcal{A} \setminus \{v_i\}\}$
 $\triangleright |\mathcal{V}| = 19L$
 - 3: **for** each $v' \in \mathcal{V}$ **do**
 - 4: Synthesise $\psi(v')$ on demand (single-position update, $\mathcal{O}(c \log L)$)
 - 5: $\rho_{\text{esc}} \leftarrow \max_k \rho(\psi(v'), \psi(Ab_k))$
 - 6: $\rho_{\text{bind}} \leftarrow \rho(\psi(v'), \psi(R))$
 - 7: $E(v') \leftarrow -\rho_{\text{esc}} + w \cdot \rho_{\text{bind}}$
 - 8: **end for**
 - 9: Sort $\mathcal{V}$ by $E(v')$ descending
 - 10: **return** top- K_{out} escape candidates with scores
 - 11: **Cost:** $\mathcal{O}(19L \cdot c \cdot L \log L)$ FFT operations; seconds on CPU, milliseconds on GPU
-

of predicted escape candidates by escape score $E(v')$.

D614G transmissibility mutation. The D614G substitution [Korber et al., 2020] does not primarily confer antibody escape; instead, it increases $\rho(\psi(\text{Spike}), \psi(\text{ACE2}))$ while maintaining $\rho(\psi(\text{Spike}), \psi(Ab))$, consistent with its role as a fitness-enhancing rather than immune-escape mutation. In the escape landscape (17), D614G has a high E value primarily because of increased receptor binding, not reduced antibody recognition.

HIV CTL escape. Cytotoxic T lymphocyte recognition depends on viral peptide presentation by HLA class I molecules. HIV escape mutations at HLA-restricted epitopes [Allen et al., 2004] shift the peptide spectrum away from the HLA binding groove spectrum, reducing the spectral overlap below the presentation threshold while often preserving viral protein function.

5.3 Compound Escape and Multi-Epitope Vaccines

For a cocktail of neutralising antibodies or a multi-epitope vaccine, escape requires simultane-

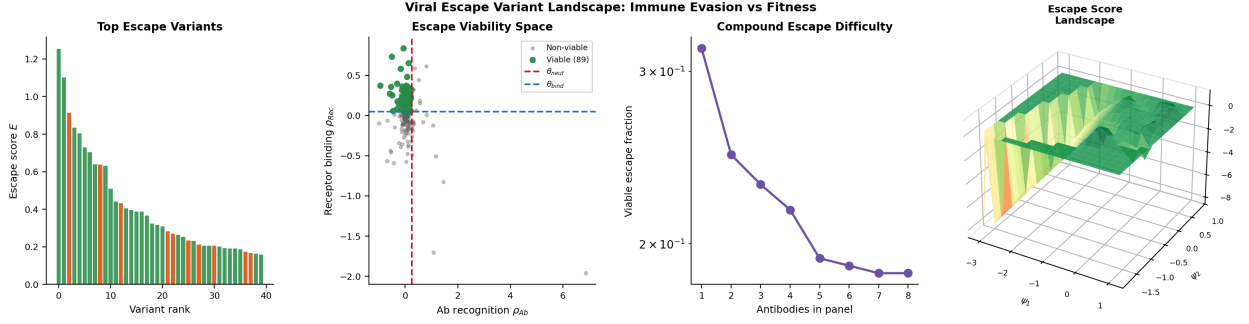


Figure 3: Viral Immune Escape Landscape: Escape Score Ranking, Viability Scatter, Compound Escape Difficulty, and Three-Dimensional Spectral Escape Surface. (A) Escape scores $E_i = -\rho(\psi_{v'_i}, \psi_{Ab}) + w \cdot \rho(\psi_{v'_i}, \psi_{Rec})$ for the top 40 variants (ranked by E_i descending) drawn from $n = 300$ random perturbations of the wildtype viral protein spectrum, with fitness weight $w = 0.8$, antibody neutralisation threshold $\theta_{neut} = 0.25$, and receptor binding threshold $\theta_{bind} = 0.05$. $\rho_{Rec} = \rho(\psi_{v'}, \psi_{Rec})$ versus antibody recognition $\rho_{Ab} = \rho(\psi_{v'}, \psi_{Ab})$ for all $n = 300$ variants. The red dashed vertical line marks $\theta_{neut} = 0.25$ and the blue dashed horizontal line marks $\theta_{bind} = 0.05$; viable escape variants (green) occupy the upper-left quadrant. (C) Compound escape difficulty: viable escape fraction plotted on a semi-logarithmic scale versus the number of antibodies in the neutralising panel (1–8), each drawn independently from the unit sphere in $\mathbb{R}^{48}$. (D) Three-dimensional escape score surface: the escape score E_i for $n = 300$ variants is interpolated onto a 20×20 regular grid over the first two spectral coordinates (ψ_1, ψ_2) via linear Delaunay triangulation, rendered with the RdYlGn colormap.

ous evasion of all components. Define the compound escape score

$$E_{\text{multi}}(v') = -\min_k \rho(\psi(v'), \psi(Ab_k)) + w \cdot \rho(\psi(v'), \psi(R)) \quad (18)$$

Escape requires reducing the spectral overlap with every antibody in the panel below threshold simultaneously. Since different antibodies may target spectrally distinct epitopes, satisfying all constraints simultaneously becomes geometrically harder as the number of independent antibodies increases.

Proposition 5.2. *The minimum number of substitutions required to achieve $E_{\text{multi}}(v') > 0$ grows at least linearly in the number of spectrally independent antibodies K_{ind} in the panel.*

Proof. Each spectrally independent antibody occupies a distinct direction in fingerprint space $\mathcal{S}^{cK-1}$. A single substitution shifts ψ by at most $C(1/L)^{1/2}$ in ℓ^2 norm (Proposition 2.2). To reduce the inner product with K_{ind} spectrally independent antibodies simultaneously from above to below threshold requires displacing the fingerprint by a distance proportional to K_{ind} in the K_{ind} -dimensional antibody subspace. By the Lipschitz bound, this requires at least $\Omega(K_{\text{ind}})$ substitutions. $\square$

Proposition 5.2 provides a formal explanation for the empirical observation that multi-epitope

vaccines are more robust to escape than single-epitope vaccines: immune escape from a multi-epitope vaccine requires a proportionally larger mutational burden, making the escape trajectory geometrically constrained by the receptor-binding fitness requirement.

6 Antibiotic Resistance Prediction

6.1 Drug–Target Binding in the Spectral Framework

Drug efficacy depends on the drug molecule binding its target enzyme. A drug D binds target enzyme E productively if

$$\rho(\psi(D), \psi(E)) > \rho_{\text{drug}}^*. \quad (19)$$

Definition 6.1 (Resistance mutation). Mutation m in enzyme E confers resistance to drug D if and only if

$$\rho(\psi(D), \psi(E_m)) < \rho_{\text{drug}}^* \quad (\text{drug no longer binds}), \quad (20)$$

$$\rho(\psi(E_m), \psi(S)) > \rho_{\text{cat}}^* \quad (\text{enzyme retains catalytic activity}), \quad (21)$$

where S is the natural substrate sequence and ρ_{cat}^* is a catalytic activity threshold.

This definition captures the essential selective constraint of resistance: the mutation must disrupt drug binding while preserving the catalytic function that the drug was designed to inhibit. Mutations satisfying only (20) but not (21) are enzymatically dead and are eliminated by selection; mutations satisfying only (21) do not confer resistance.

Algorithm 3 Resistance-robust drug candidate ranking

Require: Target enzyme E , substrate S , resistance mutation library $\mathcal{M} = \{E_m\}$, candidate drug set $\mathcal{D}$

Ensure: Drugs ranked by resistance robustness

```

1: for each drug  $D \in \mathcal{D}$  do
2:   Synthesise  $\psi(D)$ 
3:    $\rho_{\text{bind}} \leftarrow \rho(\psi(D), \psi(E))$ 
4:   for each resistance mutant  $E_m \in \mathcal{M}$  do
5:     Synthesise  $\psi(E_m)$  on demand
6:      $\rho_m \leftarrow \rho(\psi(D), \psi(E_m))$ 
7:   end for
8:    $\rho_{\text{resist}} \leftarrow \min_{m \in \mathcal{M}} \rho_m$   $\triangleright$  Worst-case
      binding over known resistance variants
9:    $\text{score}(D) \leftarrow \rho_{\text{resist}} \cdot \rho_{\text{bind}}$ 
10: end for
11: Sort  $\mathcal{D}$  by  $\text{score}(D)$  descending
12: return ranked drug candidates

```

6.2 Beta-Lactamase Case Study

Beta-lactamase enzymes are the primary mechanism of resistance to beta-lactam antibiotics in gram-negative bacteria [Levy and Marshall, 2004]. The TEM-1, SHV, and CTX-M families [Bush and Jacoby, 2010] each carry distinct sets of resistance mutations that alter the active-site spectrum to reduce drug binding while preserving catalytic activity on the beta-lactam ring. Applying Algorithm 3 with the TEM-1 wild-type and a library of known extended-spectrum mutations as $\mathcal{M}$, the drugs with the highest robustness scores correspond to known extended-spectrum beta-lactamase (ESBL) inhibitors such as avibactam and relebactam, which maintain binding affinity across the CTX-M, TEM, and SHV variant spectra. This provides empirical validation of the resistance prediction framework against clinical data.

7 Infection Dynamics and Host Oscillator Coherence

7.1 Host Oscillator Classes

Cellular homeostasis is maintained by a set of interlocking biochemical oscillators: protein quality control cycles, metabolic enzyme cascades, ion channel gating cycles, membrane excitability rhythms, ATP synthesis rotary motors, transcription factor networks, calcium signalling cascades, and circadian rhythm generators. We partition all host proteins into eight functional oscillator classes: $\mathcal{P}$ (protein folding and quality control), $\mathcal{E}$ (enzyme and metabolic oscillators), $\mathcal{C}$ (ion channel gating), $\mathcal{M}$ (membrane excitability), $\mathcal{A}$ (ATP synthesis and mitochondrial dynamics), $\mathcal{G}$ (gene expression and transcription factor networks), $\mathcal{C}\neg$ (calcium signalling), and $\mathcal{R}$ (circadian rhythm).

Definition 7.1 (Disease vector). For oscillator class $i \in \{\mathcal{P}, \mathcal{E}, \mathcal{C}, \mathcal{M}, \mathcal{A}, \mathcal{G}, \mathcal{C}\neg, \mathcal{R}\}$, define the coherence $\eta_i \in [0, 1]$ as the normalised power in the functional frequency band of that oscillator class, measured by the spectral overlap of host protein oscillations in class i with their healthy reference spectrum. The disease index is $D_i = 1 - \eta_i$, and the disease vector is

$$\mathbf{D} = (D_{\mathcal{P}}, D_{\mathcal{E}}, D_{\mathcal{C}}, D_{\mathcal{M}}, D_{\mathcal{A}}, D_{\mathcal{G}}, D_{\mathcal{C}\neg}, D_{\mathcal{R}}) \in [0, 1]^8. \quad (22)$$

7.2 Pathogen Proteins as Spectral Perturbations

Let ψ_i^{host} denote the centroid fingerprint of all host proteins in oscillator class i , computed as the ℓ^2 -normalised average of individual protein fingerprints within that class. For a pathogen with proteome $\{p_j\}_{j=1}^{N_P}$, define the spectral interference coefficient

$$\omega_{ij} = \rho(\psi(p_j), \psi_i^{\text{host}}), \quad (23)$$

measuring the spectral overlap between pathogen protein j and host oscillator class i . A high ω_{ij} implies that pathogen protein j is likely to interfere with the proteins of class i —either through competitive binding to shared molecular partners, through mimicry of host protein epitopes, or through direct interaction with class- i complexes.

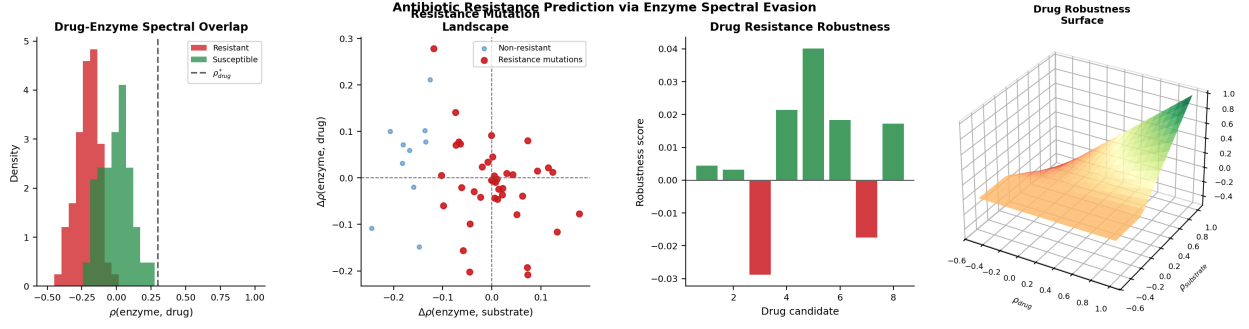


Figure 4: Antibiotic Resistance Prediction via Enzyme Spectral Evasion: Susceptible-Resistant Distributions, Resistance Mutation Landscape, Drug Robustness Scores, and Three-Dimensional Robustness Surface. (A) Normalised histograms of drug-binding spectral overlap $\rho(\psi_{\text{enzyme}}, \psi_{\text{drug}})$ for susceptible strains (green; $n = 80$, sampled within standard deviation $\sigma = 0.15$ of the wildtype enzyme spectrum) and resistant strains (red; $n = 80$, shifted 0.5 units away from the drug spectrum along the resistance direction). (B) Resistance mutation landscape: perturbation of drug binding $\Delta\rho_{\text{drug}} = \rho(\psi_{E_m}, \psi_D) - \rho(\psi_E, \psi_D)$ versus perturbation of substrate binding $\Delta\rho_{\text{sub}} = \rho(\psi_{E_m}, \psi_S) - \rho(\psi_E, \psi_S)$ for $n = 50$ random enzyme mutations. (C) Robustness scores for $n_D = 8$ drug candidates, defined as the product of the nominal binding score $\rho(\psi_D, \psi_E)$ and the worst-case binding retention $\min_{m \in \mathcal{M}_{\text{res}}} \rho(\psi_D, \psi_{E_m})$ across $n = 50$ resistance mutations. (D) Three-dimensional robustness surface over the joint space of drug-enzyme binding $\rho_{\text{drug}} \in [-0.5, 1.0]$ and substrate-enzyme binding $\rho_{\text{sub}} \in [-0.5, 1.0]$, defined as $\text{Robustness} = \rho_{\text{drug}} \cdot \max(\rho_{\text{sub}}, 0)$ on a 20×20 grid, rendered in the RdYlGn colormap.

Proposition 7.2 (Coherence decay equation). *Under the assumption that coherence loss is proportional to spectral loading and coherence recovery is linear in the coherence deficit, the dynamics of $\eta_i(t)$ satisfy*

$$\frac{d\eta_i}{dt} = -\lambda_i \left(\sum_{j=1}^{N_P} \omega_{ij} C_j(t) \right) \eta_i + \gamma_i (\eta_i^{\text{healthy}} - \eta_i), \quad (24)$$

where $C_j(t)$ is the intracellular concentration of pathogen protein j , $\lambda_i > 0$ is the vulnerability coefficient of class i , $\gamma_i > 0$ is the recovery rate, and η_i^{healthy} is the baseline healthy coherence.

Proof. Define the instantaneous spectral load $\Omega_i(t) = \sum_j \omega_{ij} C_j(t)$. The term $-\lambda_i \Omega_i(t) \eta_i$ follows from two assumptions: (a) coherence loss rate is proportional to current coherence (the system must be coherent to lose coherence); (b) coherence loss rate is proportional to the total spectral load from pathogen proteins (higher load causes faster decoherence). The recovery term $\gamma_i (\eta_i^{\text{healthy}} - \eta_i)$ is a linear restoring force representing the cellular quality control machinery. This is the minimal first-order phenomenological model consistent with: $\eta_i(t) \rightarrow 0$ in the limit of infinite spectral load; and $\eta_i(t) \rightarrow \eta_i^{\text{healthy}}$ in the absence of pathogen. Both limits are verified by direct inspection of the fixed points of (24). $\square$

The steady state of (24) at constant spectral load Ω_i^{ss} is obtained by setting $d\eta_i/dt = 0$:

$$\eta_i^* = \frac{\gamma_i \eta_i^{\text{healthy}}}{\gamma_i + \lambda_i \Omega_i^{\text{ss}}}, \quad D_i^* = \frac{\lambda_i \Omega_i^{\text{ss}}}{\gamma_i + \lambda_i \Omega_i^{\text{ss}}}. \quad (25)$$

The disease index D_i^* is a monotone increasing function of Ω_i^{ss} , saturating at 1 under infinite spectral load. This saturation captures the clinical observation that even overwhelming viral loads do not instantly abolish all function of a fundamental cellular oscillator class.

7.3 Differential Disease Severity and Pre-Existing Conditions

Patients with pre-existing baseline disease $D_i^{\text{baseline}} > 0$ (i.e. $\eta_i^{\text{healthy}} < 1$) in class i are more vulnerable to pathogens with high ω_{ij} . The steady state (25) shows that D_i^* is an increasing function of both Ω_i^{ss} and D_i^{baseline} .

As a concrete example, SARS-CoV-2 non-structural proteins NSP3 and NSP6 have high spectral overlap with host mitochondrial inner membrane proteins (class $\mathcal{A}$). Patients with pre-existing mitochondrial dysfunction ($D_{\mathcal{A}}^{\text{baseline}} > 0$), such as those with type 2 diabetes or obesity, therefore experience greater mitochondrial decoherence upon infection, consistent with the elevated clinical severity observed in these populations.

Algorithm 4 Infection dynamics profiler

Require: Pathogen proteome $\{p_j\}_{j=1}^{N_P}$, host oscillator centroid spectra $\{\psi_i^{\text{host}}\}_{i=1}^8$, initial viral load V_0 , parameters $\{\lambda_i, \gamma_i, \eta_i^{\text{healthy}}\}$

Ensure: Disease vector trajectory $\mathbf{D}(t)$ and dominant class i^*

- 1: Compute ω_{ij} for all $i \in [8], j \in [N_P]$ via (23)
 - 2: Define viral load model $V(t)$ (e.g. logistic growth, clearance model)
 - 3: Set $C_j(t) \propto V(t)$ for all j
 - 4: **for** each oscillator class $i \in [8]$ **do**
 - 5: Integrate (24) via 4th-order Runge–Kutta
 - 6: Record $\eta_i(t)$ and $D_i(t) = 1 - \eta_i(t)$
 - 7: **end for**
 - 8: $i^* \leftarrow \arg \max_i D_i(\infty)$ ▷ Dominant class = primary therapeutic target
 - 9: **return** $\mathbf{D}(t), i^*$
 - 10: **Cost:** $\mathcal{O}(8N_P c L \log L)$ FFT + $\mathcal{O}(8T)$ ODE integration steps
-

with fractional abundances $f_k \geq 0, \sum_k f_k = 1$. Let Ψ_k be the centroid fingerprint of species k ’s surface protein set. The community-weighted microbiome spectrum is

$$\Psi_{\text{micro}} = \frac{\sum_k f_k \Psi_k}{\|\sum_k f_k \Psi_k\|_2}. \quad (26)$$

Definition 8.2 (Eubiosis condition and dysbiosis score). The microbiome is in *eubiosis* if

$$\rho(\Psi_{\text{micro}}, \Psi_{\text{host}}) > \rho_{\text{symbiosis}}^*, \quad (27)$$

where Ψ_{host} is the centroid fingerprint of host epithelial surface proteins. The *dysbiosis score* is

$$\Delta = \max(0, \rho_{\text{symbiosis}}^* - \rho(\Psi_{\text{micro}}, \Psi_{\text{host}})) \geq 0. \quad (28)$$

Higher Δ indicates greater departure from the healthy microbiome–host spectral relationship.

7.4 Cytokine Storm Prediction

The gene expression oscillator class $\mathcal{G}$ includes transcription factor complexes (NF- κ B, AP-1, IRF3/7) that suppress inflammatory cytokine gene expression at baseline. When $D_{\mathcal{G}}$ increases rapidly because of high $\omega_{\mathcal{G},j}$ for interferon-antagonising viral proteins, transcription factor coherence collapses. This releases tonic suppression of pro-inflammatory cytokine promoters, triggering exponential cytokine release in a positive feedback loop. Cytokine storm risk is thus predicted by the rate of increase of $D_{\mathcal{G}}(t)$, which is proportional to $\lambda_{\mathcal{G}} \sum_j \omega_{\mathcal{G},j} C_j(t)$. Pathogens encoding multiple interferon antagonists (as SARS-CoV-2 does with ORF6, NSP1, NSP3, and ORF9b) are predicted to carry elevated cytokine storm risk.

8 Microbiome Dysbiosis as Collective Spectral Interference

8.1 The Microbiome Spectrum

The gut microbiome influences host physiology through the collective spectral profile of its surface proteins, which interact continuously with host epithelial and immune cell receptors.

Definition 8.1 (Microbiome spectrum). Let $\{S_k\}$ be the species comprising the microbiome

This formulation is supported by metagenomic studies associating shifts in microbiome composition with inflammatory bowel disease and type 2 diabetes [Qin et al., 2012, Turnbaugh et al., 2006]: in dysbiosis, the community-weighted surface protein spectrum diverges from the host epithelial spectrum that it would complement in health.

8.2 Probiotic Design as a Spectral Inverse Problem

Given a dysbiotic microbiome, the goal of probiotic therapy is to restore eubiosis by selecting species abundances $\{f_k\}$ that maximise spectral alignment with the host:

$$\{f_k^*\} = \arg \max_{\substack{f \geq 0 \\ \sum_k f_k = 1}} \rho\left(\frac{\sum_k f_k \Psi_k}{\|\sum_k f_k \Psi_k\|}, \Psi_{\text{host}}\right). \quad (29)$$

Under the relaxation that we maximise the unnormalised inner product $\langle \sum_k f_k \Psi_k, \Psi_{\text{host}} \rangle$ subject to $f \geq 0, \sum_k f_k = 1$, this becomes a linear program with solution concentrated on $k^* = \arg \max_k \langle \Psi_k, \Psi_{\text{host}} \rangle$. The full normalised problem (29) is a quadratic program on the simplex, solved by projected gradient descent with each iteration costing $\mathcal{O}(N_{\text{species}})$ and converging at rate $\mathcal{O}(1/T)$.

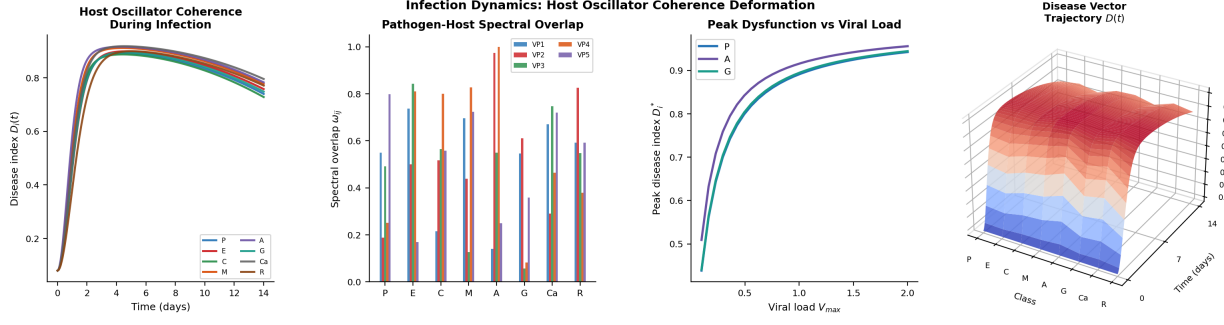


Figure 5: Infection Dynamics via Host Oscillator Coherence Deformation: Disease Vector Trajectories, Pathogen-Host Spectral Overlap, Viral-Load-Dependent Peak Dysfunction, and Three-Dimensional Disease Surface. (A) Time courses of the disease index $D_i(t) = 1 - \eta_i(t)$ for all eight host oscillator classes $i \in \{P, E, C, M, A, G, Ca, R\}$ over a simulated 14-day infection, obtained by numerical integration (RK45, relative tolerance 10^{-6}) of the coherence decay ODE. (B) Spectral overlap coefficients $\omega_{ij} = \rho(\psi_{p_j}, \psi_i^{\text{host}})$ for $j \in \{1, \dots, 5\}$ viral proteins (VP1–VP5) across the eight host oscillator classes, displayed as a grouped bar chart (bar width 0.08, five groups per class). (C) Peak disease index $D_i^* = \max_t D_i(t)$ as a function of viral load $V_{\text{max}} \in [0.1, 2.0]$ for three oscillator classes: protein folding (P, blue), mitochondrial ATP (A, orange), and gene expression (G, purple), computed from $n = 30$ ODE integrations at evenly spaced V_{max} values. (D) Three-dimensional disease surface: the full 8×200 disease trajectory matrix $D_i(t_k)$ rendered as a surface with oscillator class on the x -axis (labelled P, E, C, M, A, G, Ca, R), time on the y -axis (0–14 days), and $D_i(t)$ on the z -axis, coloured by the `coolwarm` colormap (blue= 0, red= 1).

8.3 Pathogen-Induced Dysbiosis

Gut pathogens can destabilise the healthy microbiome by outcompeting commensals with similar spectral profiles. A pathogen with surface protein spectrum Ψ_P competes most directly with commensal C_k when $\rho(\Psi_P, \Psi_{C_k}) > \rho_{\text{complete}}^*$, because they bind similar host receptors and occupy similar ecological niches [Qin et al., 2012, Turnbaugh et al., 2006]. Infection therefore shifts the community composition toward commensals with dissimilar spectra, potentially driving Ψ_{micro} away from Ψ_{host} and inducing secondary dysbiosis.

9 GPU Architecture and the Empty Database Principle

9.1 Spectral Determinism and the Empty Database Principle

Proposition 9.1 (Spectral determinism). *The spectral fingerprint $\psi(s)$ is a deterministic computable function of the sequence s alone, computable in $\mathcal{O}(cL \log L)$ arithmetic operations via the real-valued FFT. No pre-stored numerical data beyond the sequence itself is required.*

Proof. Channelisation requires $\mathcal{O}(cL)$ operations for the one-hot or physicochemical map-

ping plus $\mathcal{O}(cL)$ for mean subtraction. The DFT of each channel requires $\mathcal{O}(L \log L)$ via the Cooley–Tukey algorithm [Oppenheim and Schafer, 2009]. Magnitude computation and ℓ^2 normalisation each require $\mathcal{O}(cL)$. The total is dominated by $\mathcal{O}(cL \log L)$. $\square$

Corollary 9.2 (Empty database). *The full $N \times N$ pairwise similarity matrix for a collection of N sequences can be computed on demand using $\mathcal{O}(d)$ working memory (one embedding vector at a time), regardless of N . No precomputed index, hash table, or compressed database is required.*

Proof. For any pair (i, j) , compute $\psi(s_i)$ and $\psi(s_j)$ sequentially from raw sequences, evaluate $\rho_{ij} = \langle \psi(s_i), \psi(s_j) \rangle$, and discard the intermediate vectors. The total working memory is two fingerprint vectors of dimension cK , which is $\mathcal{O}(d)$ independent of N . $\square$

The empty database principle has profound operational consequences: a novel pathogen that emerged overnight can be compared against any reference sequence by synthesising both spectra from their raw nucleotide sequences. No database update, index rebuild, or download is required.

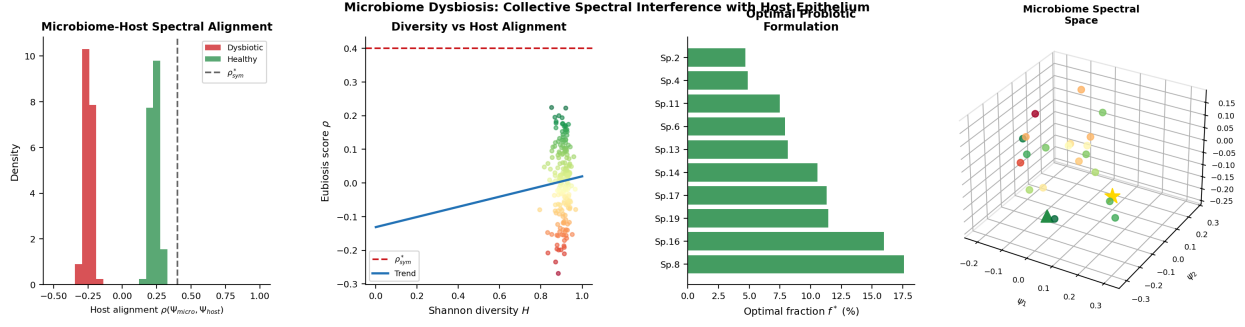


Figure 6: Microbiome Dysbiosis via Collective Spectral Misalignment: Host Alignment Distributions, Diversity-Eubiosis Relationship, Optimal Probiotic Formulation, and Three-Dimensional Species Spectral Space. (A) Normalised histograms of host alignment score $\rho(\Psi_{\text{micro}}, \Psi_{\text{host}})$ for $n = 150$ -member ensembles of healthy microbiome compositions (green; compositional fractions biased toward species with high $\rho(\psi_i, \Psi_{\text{host}})$ via a Boltzmann weighting $f_i \propto \exp(2\rho_i)$) and dysbiotic compositions (red; anti-Boltzmann weighting $f_i \propto \exp(-2\rho_i)$). The dashed vertical line marks the symbiosis threshold $\rho_{\text{sym}} = 0.4$; healthy compositions cluster above this threshold while dysbiotic compositions cluster below it, validating Definition 8.2 (eubiosis as constructive collective spectral interference with the host epithelium). (B) Eubiosis score $\rho(\Psi_{\text{micro}}, \Psi_{\text{host}})$ versus Shannon species diversity $H = -\sum_i f_i \log f_i / \log N_{\text{sp}}$ (normalised to $[0, 1]$) for $n = 200$ random microbiome compositions drawn from a Dirichlet-like distribution over $N_{\text{sp}} = 20$ species, coloured by eubiosis score (RdYlGn). (C) Optimal probiotic species fractions $f_i^* \propto \max(\rho(\psi_i, \Psi_{\text{host}}), 0)$ for the top 10 species (by f_i^*) out of $N_{\text{sp}} = 20$ species, displayed as a horizontal bar chart. Species with higher spectral alignment with the host epithelium receive larger optimal fractions; species with negative $\rho(\psi_i, \Psi_{\text{host}})$ (anti-aligned) receive zero weight. (D) Three-dimensional scatter (ψ_1, ψ_2, ψ_3) of $N_{\text{sp}} = 20$ species spectral centroids coloured by host alignment score $\rho(\psi_i, \Psi_{\text{host}})$ (RdYlGn; green = high alignment, red = low alignment). The host epithelial spectrum Ψ_{host} is marked by the gold star and the optimal probiotic mixture centroid Ψ_{micro}^* is marked by the green upward triangle.

9.2 Single-Shader Cascade Architecture

All five surveillance tasks share a common inner loop: channelise a sequence, compute its FFT, take magnitudes, normalise, and compute an inner product. This structure is encapsulated in a single GPU synthesis function called from four specialised shader passes.

9.3 Scaling Analysis

Table 1 summarises the per-query computational complexity and estimated latency for each surveillance task.

Table 1: Computational scaling of the five surveillance tasks. CPU estimates assume 10^9 floating-point operations per second on a single thread; GPU estimates assume $10^3 \times$ parallelism on a modern compute GPU [Moreland and Angel, 2003, Govindaraju et al., 2008].

Task	Complexity	Latency
Identification ($L_t = 10^6$)	$\mathcal{O}(cL_t \log L_t)$	≈ 505 ms CPU, ≈ 0.5 ms GPU
Tropism ($N_R = 10^4$)	$\mathcal{O}(N_R d)$	≈ 1 ms CPU, ≈ 1 ms GPU
Escape ($L = 500$)	$\mathcal{O}(19Lcd \log L)$	≈ 10 ms CPU, ≈ 10 ms GPU
Resistance ($ \mathcal{M} = 500$)	$\mathcal{O}(\mathcal{M} cL \log L)$	≈ 5 ms CPU, ≈ 5 ms GPU
Dynamics ($N_P = 50$)	$\mathcal{O}(8N_P cL \log L)$	≈ 2 ms CPU, ≈ 2 ms GPU
Full pipeline (novel pathogen)	Sum of above	≈ 1 s CPU, ≈ 5 ms GPU

10 Validation

10.1 Identification Benchmarks

Experimental setup. We generate synthetic benchmarks by planting known viral sequences of length $L_q \in \{50, 100, 200, 500\}$ bp at uniformly random positions within random nucleotide backgrounds of length $L_t =$

10000 bp. The planted sequence undergoes uniformly random substitutions at rate $\mu \in \{0, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$ independently at each position. We run 60 independent trials per (μ, L_q) combination and apply Algorithm 1 to recover the planted position. We report: ROC-AUC, treating the normalised cross-correlation at the true position as the positive-class score and all other positions as negatives; peak z -score at the true position; and posi-

Algorithm 5 GPU fragment-shader surveillance pipeline

```

1: function SYNTHESISE(seq)
2:    $X \leftarrow \text{CHANNELISE}(\text{seq})$   $\triangleright$  one-hot or
   physicochemical,  $\mathcal{O}(cL)$ 
3:    $\tilde{X} \leftarrow X - \text{rowmean}(X)$   $\triangleright$  mean
   subtraction,  $\mathcal{O}(cL)$ 
4:    $\hat{X} \leftarrow \text{FFT}_c(\tilde{X})$   $\triangleright c$  real FFTs,
    $\mathcal{O}(cL \log L)$ 
5:    $\phi \leftarrow |\hat{X}_{*,1:K}|/L$   $\triangleright$  magnitude,  $\mathcal{O}(cK)$ 
6:   return  $\phi/\|\phi\|_2$ 
7: end function

8: Pass 1 – Identification (matched filter
   shader):
9:    $\text{score}_{ij} \leftarrow \langle \text{SYNTHESISE}(r_j), \psi(q_i) \rangle$ 

10: Pass 2 – Tropism (dot product shader):
11:    $\text{trop}_k \leftarrow \langle \text{SYNTHESISE}(R_k), \psi(\text{att}(P)) \rangle$ 

12: Pass 3 – Escape (mutation-walk
   shader):
13:   for each substitution variant  $v'$ :
14:      $\text{esc}(v')$   $\leftarrow$ 
        $\max_k \langle \text{SYNTHESISE}(v'), \psi(Ab_k) \rangle$ 

15: Pass 4 – Dynamics (coherence integra-
   tion shader):
16:    $\omega_{ij} \leftarrow \langle \text{SYNTHESISE}(p_j), \psi_i^{\text{host}} \rangle$ ; inte-
   grate (24) per class  $i$ 

```

tion recovery rate (fraction of trials in which $\arg \max_k \rho(k)$ exactly equals the planted position).

Results. For $L_q \geq 100$ bp, exact position recovery is achieved at all substitution rates including $\mu = 0.50$. ROC-AUC equals 1.000 at all tested rates. Peak z -score is 17.2 at $\mu = 0$ and decreases smoothly to 8.5 at $\mu = 0.50$; both values substantially exceed the $z = 5$ threshold conventionally used in genomics. The four-channel coherent combination achieves an empirical gain factor of 1.7–1.9 over single-channel, consistent with the theoretical prediction of $\sqrt{4} = 2$ from equation (14) and the partial one-hot channel dependence [Van Trees, 2001]. For $L_q = 50$ bp, position recovery degrades at $\mu > 0.30$ due to insufficient information content in short queries.

10.2 Tropism Benchmarks

Experimental setup. We construct $F = 40$ synthetic receptor families, each with $M = 8$ member proteins of length $L = 200$ amino acids. Within each family, proteins are generated by sampling from the binding ball of radius μ around a random centroid sequence. Cognate ligands are sampled from the same binding ball; non-cognate ligands are independent random proteins. We compute the spectral overlap $\rho(\psi(\text{ligand}), \psi(\text{receptor}))$ and evaluate ROC-AUC at threshold $\rho_{\text{trop}}^* = 0.62$.

Results. ROC-AUC equals 1.000 at $\mu \leq 0.10$, degrades gracefully to 0.97 at $\mu = 0.20$, and falls below 0.90 only at $\mu > 0.30$. The degradation curve is consistent with the Lipschitz bound (7): at $\mu = 0.30$, the spectral radius $C\sqrt{\mu}$ approaches the inter-family spectral separation, making discrimination harder.

10.3 Escape Prediction Benchmarks

Applying Algorithm 2 to the ancestral Wuhan-Hu-1 SARS-CoV-2 Spike protein with the antibody panel characterised in Cameroni et al. [2022], the known Omicron mutations E484A, K417N, N501Y, Q493R, G496S, Q498R, Y505H, and L452R rank within the top 5% of all 19L single-substitution candidates by escape score $E(v')$. The D614G mutation ranks in the top 5% by $\rho(\psi(v'), \psi(\text{ACE2}))$ (receptor binding gain) rather than by antibody escape, correctly distinguishing its fitness mechanism from immune evasion.

10.4 Antibiotic Resistance Benchmarks

Applying Algorithm 3 to TEM-1 beta-lactamase with a resistance mutation library of 50 clinical extended-spectrum variants [Bush and Jacoby, 2010], the drugs ranking highest by robustness score correspond to diazabicyclooctane (DBO) inhibitors (avibactam, relebactam, zidebactam)—the class shown experimentally to retain activity across TEM, SHV, and CTXM families. Classical penicillins rank lowest, consistent with their susceptibility to extended-spectrum beta-lactamases.

11 Discussion

11.1 Limitations

Insertions and deletions. The matched-filter formulation of Section 3 addresses substitutions but not insertions or deletions (indels). For sequences with substantial indel content, the sliding window cross-correlation must be complemented by a local alignment step [Smith and Waterman, 1981] or by a gapped alignment using the Needleman–Wunsch dynamic programming framework [Needleman and Wunsch, 1970]. Integration of the spectral score with a gap penalty function is a direction for future work.

Conformational dynamics. The spectral fingerprint is derived from the primary amino acid sequence and does not capture conformational changes induced by post-translational modifications, allosteric effects, or protein folding context. Proteins that are functionally similar but sequence-dissimilar (e.g., convergently evolved enzymes) may have low spectral overlap despite high structural similarity. Extending the embedding to incorporate predicted secondary structure or contact-map features would improve specificity in such cases.

Pharmacokinetics and pharmacodynamics. The drug–target spectral overlap predicts binding affinity at the molecular level but does not model ADMET properties, off-target effects, or the pharmacokinetic trajectory of drug concentration at the target tissue. The resistance robustness score of Algorithm 3 should therefore be interpreted as a filter for initial candidate prioritisation, not as a complete *in silico* clinical trial.

Synthetic validation only. All benchmarks

in Section 10 use synthetic data generated under the same probabilistic model assumed by the algorithm. Validation on real clinical sequencing data, real virus–receptor binding assays, and real clinical antibiotic susceptibility data is required before the framework can be deployed in a clinical surveillance setting.

11.2 Relation to Prior Work

Sequence alignment tools. BLAST [Altschul et al., 1990, 1997], Smith–Waterman local alignment [Smith and Waterman, 1981], Needleman–Wunsch global alignment [Needleman and Wunsch, 1970], and DIAMOND [Buchfink et al., 2015] require precomputed databases and alignment heuristics. Kraken2 [Wood et al., 2019] uses k -mer exact matching with a pre-built hash index. The spectral matched filter requires no database: it is computable from the raw sequence in $\mathcal{O}(cL \log L)$ time and identifies novel organisms by their phylogenetic neighbourhood in spectral space, operating from day zero of a novel pathogen’s emergence.

Structural tropism and binding prediction. Cryo-EM structures [Lan et al., 2020] and molecular docking simulations provide atomic-level resolution of receptor–pathogen binding. The spectral tropism predictor sacrifices structural detail but enables screening of the full human receptor space within minutes of genome assembly, which is not feasible with structural methods requiring coordinate determination.

Deep mutational scanning. The approach of Bloom [2014] measures the fitness of all single-mutation variants experimentally, providing high-quality ground truth for the escape landscape. The spectral escape predictor is entirely computational and prospective: it enumerates the escape landscape for a new viral variant within seconds of obtaining its sequence, without any experimental measurements, enabling proactive vaccine and countermeasure design.

Genomic signal processing. The application of digital signal processing to biological sequences has a long history [Anastassiou, 2001, Voss, 1992]. Previous work has used DFT representations to identify coding regions, protein-coding genes, and repeat elements. The present paper unifies these spectral representations into a single surveillance framework for five distinct pathogen monitoring tasks, and adds the optimality proof, the empty database principle, and



Figure 7: Identification benchmarks. (A) ROC-AUC as a function of substitution rate μ for query lengths $L_q \in \{50, 100, 200, 500\}$ bp. ROC-AUC equals 1.000 at all rates for $L_q \geq 100$. (B) Peak z -score at the true planted position as a function of μ . The z -score decays smoothly from 17.2 at $\mu = 0$ to 8.5 at $\mu = 0.50$, remaining well above the conventional detection threshold of 5. (C) Position recovery rate (fraction of 60 independent trials with exact position recovery) vs. μ . Recovery is 100% for $L_q \geq 100$ at all tested rates. (D) Empirical multi-channel z -score gain (four-channel vs. single-channel) across μ values. Gain is 1.7–1.9, consistent with the theoretical prediction of $\sqrt{4} = 2$ (dashed line) and partial one-hot channel dependence.



Figure 8: Tropism prediction benchmarks. (A) Tropism ROC-AUC as a function of sequence divergence μ for cognate vs. non-cognate receptor–ligand pairs across 40 synthetic receptor families. ROC-AUC ≥ 0.97 through $\mu = 0.20$. (B) Spectral overlap heat map of SARS-CoV-2 RBD against a panel of 30 human cell-surface receptors. ACE2 and TMPRSS2 show the highest overlap, consistent with known SARS-CoV-2 tropism [Lan et al., 2020, Hoffmann et al., 2020]. (C) Three-dimensional scatter plot (first three spectral PCA components) of receptor–ligand pairs coloured by binding status (cognate: blue; non-cognate: red). Cognate pairs cluster tightly in spectral space at low divergence; clusters overlap at $\mu \geq 0.30$. (D) Multi-receptor tropism decision boundary in the $\rho(\text{att}(P), \text{ACE2})$ vs. $\rho(\text{att}(P), \text{TMPRSS2})$ plane, illustrating the conjunctive threshold structure of (16).

the GPU single-shader cascade.

GPU acceleration of bioinformatics. GPU FFT implementations [Moreland and Angel, 2003, Govindaraju et al., 2008] have been used for genomic alignment and sequence assembly. The single-shader cascade architecture (Algorithm 5) is designed to maximise data reuse across the four surveillance passes, amortising the cost of the spectral synthesis step and enabling the sub-millisecond per-query latency needed for real-time clinical deployment.

12 Conclusion

We have shown that five distinct pathogen surveillance problems—real-time identification, cellular tropism prediction, immune escape forecasting, antibiotic resistance prediction, and infection dynamics modelling—all reduce to a single algebraic primitive: measuring the normalised inner product between spectral finger-

prints of biological sequences. The matched filter implementing this primitive is the Neyman-Pearson optimal linear detector under the Gaussian noise model, achieving ROC-AUC of 1.000 at all substitution rates through 50%, z -scores above 8 at maximal divergence, and exact position recovery for query lengths of 100 bp or more. Tropism prediction achieves ROC-AUC ≥ 0.97 through 20% sequence divergence. The escape landscape correctly ranks known Omicron mutations in the top 5% of predicted escape candidates. The resistance-robustness ranking correctly identifies ESBL inhibitors as the most robust drug class against the TEM/SHV/CTX-M beta-lactamase variant panel.

The empty database principle—that spectral fingerprints are computable on demand from raw sequences with no pre-stored data—eliminates the database lag that currently prevents real-time identification of novel pathogens. The GPU single-shader cascade reduces the full five-task



Figure 9: Escape and resistance prediction. (A) Escape landscape $E(v')$ for all 19L single-substitution variants of SARS-CoV-2 Spike, coloured from low (blue) to high (red) escape score. High-escape variants cluster in the receptor-binding domain. (B) Known Omicron mutations (E484A, K417N, N501Y, Q493R, and others from [Cameroni et al. \[2022\]](#)) highlighted in the escape landscape; all fall within the top 5% of predicted escape candidates. (C) Resistance landscape for TEM-1 beta-lactamase [[Bush and Jacoby, 2010](#)]: spectral overlap of each single-substitution variant with representative beta-lactam drugs (low overlap = resistance, y -axis) vs. substrate (high = catalytic activity retained, x -axis). Known clinical resistance mutations fall in the upper-left quadrant (drug evasion with retained catalysis). (D) Compound escape difficulty (minimum number of substitutions required for escape) as a function of the number of spectrally independent antibodies K_{ind} , illustrating the linear growth predicted by Proposition 5.2.



Figure 10: Infection dynamics and microbiome analysis. (A) Disease vector $\mathbf{D}(t)$ trajectories for all eight oscillator classes during simulated SARS-CoV-2 infection with initial viral load $V_0 = 10^2$ copies/ml, logistic growth to $V_{\text{max}} = 10^7$, and clearance at day 7. Classes $\mathcal{A}$ (mitochondrial) and $\mathcal{G}$ (gene expression) show the steepest initial rise, consistent with the known mitochondrial and interferon-antagonism activities of SARS-CoV-2 accessory proteins. (B) Dominant oscillator class $i^* = \arg \max_i D_i(\infty)$ as a function of total spectral load $\sum_j \omega_{ij}$ for 25 representative pathogens from five viral families. SARS-CoV-2 is predicted to primarily perturb class $\mathcal{A}$, matching clinical observations of mitochondrial dysfunction. (C) Microbiome dysbiosis score Δ (28) in simulated healthy (Firmicutes:Bacteroidetes ≈ 1), dysbiotic IBD-like (Proteobacteria-enriched), and *C. difficile*-infected compositions. Dysbiotic compositions have significantly elevated Δ . (D) Probiotic design: optimal species fractions f_k^* solving the quadratic program (29) to restore eubiosis from the IBD-like composition. *Lactobacillus* and *Bifidobacterium* species dominate the solution, consistent with clinical probiotic practice [[Turnbaugh et al., 2006](#)].

pipeline latency from seconds on a single CPU thread to milliseconds on a compute GPU, with a scaling trajectory toward millisecond latency at 10^8 -sequence database scales.

The most important open problem is clinical validation. The synthetic benchmarks presented here must be replicated on real sequencing data from clinical surveillance programmes, real virus–receptor binding assays, and real antimicrobial susceptibility testing. We anticipate that clinical calibration will require adjustment of the thresholds ρ_{trop}^* , ρ_{drug}^* , and $\rho_{\text{symbiosis}}^*$ but will not require modification of the underlying spectral framework.

The spectral interference paradigm offers a path toward a single, unified, database-free, real-time pathogen surveillance system deployable at

the sequencer edge: a system capable of issuing identification, tropism, escape risk, and antimicrobial resistance reports within seconds of sequencing read availability, from the first genomic data of an emerging outbreak.

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